

PHUOPSIS

hardy. Good for groundcover, especially on banks and in rock gardens. Needs sun and well-drained soil. Propagate by division in spring, by semiripe cuttings in summer, or by seed sown in fall. ***P. stylosa***, syn. *Crucianella stylosa*, illus. p.338.

PHYGELIUS

SCROPHULARIACEAE

Genus of evergreen or semievergreen shrubs and subshrubs, grown for their showy, tubular flowers. Frost hardy, but in most areas plant in a sheltered site; will attain a considerably greater height when grown against a warm, sunny wall. Needs sun and fertile, well-drained, but not too dry, soil. Usually loses leaves or has shoots cut to ground by frosts. Cut back to just above ground level in spring, or, if plants have woody bases, prune to live wood. Propagate by softwood cuttings in summer.

P. aequalis illus. p.156. **‘Yellow Trumpet’** illus. p.160. ***P. capensis*** (Cape figwort). Evergreen or semievergreen, upright shrub. H 4ft (1.2m), S 5ft (1.5). Has triangular, dark green leaves. Tall, slender spires bear tubular, bright orange-red flowers, turning back toward the stems, each with a red mouth and a yellow throat, from midsummer to early fall. Z5–9 ***P. x rectus*** **‘Winchester Fanfare’**. Evergreen or semievergreen, upright shrub. H and S 5ft (1.5). Has triangular, dark green leaves. Produces tubular, nodding, pale red flowers, each with scarlet lobes and a yellow throat, from midsummer to early fall. Z5–9

Phyllanthus nivosus. See *Breynia disticha*.

X PHYLLIOPSIS

ERICACEAE

Hybrid genus (*Phyllodoce x almiopsis*) of dwarf, evergreen shrubs, grown for their bell-shaped flowers. Suitable for peat beds or rock gardens. Fully hardy. Needs partial shade and peaty, acidic soil. Trim back after flowering to keep compact. Propagate by semiripe cuttings in late summer. ***x P. hillieri*** **‘Pinocchio’**. Evergreen, upright shrub. H 8in (20cm), S 10in (25cm). Branched stems bear thin, ovate leaves. Long, open clusters of bell-shaped, very deep pink flowers are produced in spring and intermittently thereafter. Z3–8

Phyllitis scolopendrium. See *Asplenium scolopendrium*.

PHYLLOCLADUS

PODOCARPACEAE

See also CONIFERS. ***P. aspleniifolius*** (Tasman celery pine). Slow-growing, upright conifer. H 15–30ft (5–10m), S 10–15ft (3–5m). Instead of true leaves has flattened, leaflike shoots known as phylloclades; these are dull dark green and resemble celery leaves in outline. Produces inedible, white-coated nuts with fleshy, red bases. Z10–11 ***P. trichomanoides*** illus. p.101.

PHYLLODOCE

ERICACEAE

Genus of evergreen shrubs and subshrubs, grown for their heathlike, leathery leaves, and their attractive flowers. Fully hardy. Requires partial shade and moist, peaty, acidic soil. Propagate by semiripe cuttings in late summer or by seed in spring. ***P. caerulea***, syn. *P. taxifolia*, illus. p.334. ***P. empetriformis*** illus. p.334. ***P. x intermedia*** **‘Drummondii’** illus. p.333. **‘Fred Stoker’** is an evergreen, upright subshrub. H and S 9in (23cm). Has narrow, glossy, fine-toothed, dark green leaves. From late spring to early summer, bears clusters of pitcher-shaped, bright reddish-purple flowers, on slender, red stalks, at the stem tips. Z7–10 ***P. nipponica***. Evergreen, upright, freely branched subshrub. H and S to 8in (20cm). Has fine, linear, dark green leaves, with white-downy midribs and rolled margins. In late spring and early summer, loose clusters of stalked, pendant, bell-shaped, white flowers are borne from stem tips on upright, red-tinted stalks. Z6–10 ***P. taxifolia***. See *P. caerulea*.

PHYLLOSTACHYS

POACEAE/GRAMINEAE

See also GRASSES, BAMBOOS, RUSHES, and SEDGES. ***P. aurea*** (Golden bamboo) illus. p.288. ***P. aureosulcata*** (Yellow-groove bamboo). Evergreen, clump-forming bamboo. H 10–20ft (3–6m), S indefinite. Has striped sheaths and yellow grooves on rough, brownish-green stems. Mid-green leaves are up to 6in (15cm) long; rarely produces flowers. Z6–10 ***P. bambusoides*** (Timber bamboo) illus. p.287. ***P. flexuosa*** (Zigzag bamboo). Evergreen, clump-forming bamboo. H 6–30ft (2–10m), S indefinite. Has slender, markedly zigzag stems that turn black with age. Leaf sheaths have no bristles. Leaves stay fresh green all winter. Z6–10 ***P. ‘Henonis’***. See *P. nigra* f. *henonis*. ***P. nigra*** (Black bamboo) illus. p.286. **f. henonis** (syn. *P. ‘Henonis’*) illus. p.287. ***P. viridiglaucescens*** illus. p.287. ***P. vivax* f. aureocaulis** illus. p.289.

X PHYLLOTHAMNUS

ERICACEAE

Hybrid genus (*Phyllodoce x Rhodothamnus*) of one species of evergreen shrub, grown for its heathlike, linear leaves and its funnel-shaped flowers. Good for a peat bed or shady rock garden. Fully hardy. Needs a sheltered, partially shaded site and moist, acidic soil. Propagate by semiripe cuttings in late summer. ***x P. erectus***. Evergreen, upright shrub. H 10in (25cm), S 1ft (30cm). Has small, linear, glossy, deep green leaves. Clusters of slender-stalked, widely funnel-shaped, soft rose-pink flowers, ½in (1cm) across, are borne in late spring and early summer. Z3–8

Phyodina. See *Callisia*.

PHYSALIS

Chinese lantern

SOLANACEAE

Genus of one variable, summer-flowering perennial, grown for its decorative, lanternlike calyces and fruits. Fully to half hardy. Grows in sun or shade and in well-drained soil. Propagate by division or softwood cuttings in spring, annuals by seed in spring or fall. ① All parts of *P. alkekengi*, except the fully ripe fruit, may cause mild stomach upset if ingested; contact with foliage may irritate skin. ***P. alkekengi*** (Bladder cherry, Winter cherry). Spreading perennial, grown as an annual. H 2–2½ft (60–75cm), S 3ft (90cm) or more. Leaves are mid-green and ovate. Inconspicuous, nodding, star-shaped, white flowers in summer are followed, in fall, by rounded, bright orange-red fruits, surrounded by inflated, orange calyces. Z6–9

PHYSOCARPUS

ROSACEAE

Genus of deciduous, mainly summer-flowering shrubs, grown for their foliage and flowers. Fully hardy. Requires sun and fertile, not too dry soil. Prefers acidic soil and does not grow well on shallow, chalky soil. Thin established plants occasionally by cutting some older shoots back to ground level after flowering. Propagate by softwood cuttings in summer. ***P. opulifolius*** (Ninebark). Deciduous, arching, dense shrub. H 6–10ft (2–3m). Has peeling bark and broadly ovate, toothed and lobed, mid-green leaves. Bears clusters of tiny, at times pink-tinged, white flowers in early summer. Z2–8. **‘Dart’s Gold’** illus. p.159. **‘Diabolo’** has dark blackish-red leaves. **LADY IN RED (‘Tuildad’)**, H 6ft (2m), S 4ft (1.2m), is compact and tightly branched, with copper-red foliage and pale pink flowers.

PHYSOPLEXIS

CAMPANULACEAE

Genus of one species of tufted perennial, grown for its flowers. Good grown on tufa, in rock gardens, troughs, and alpine houses. Fully hardy. Needs sun and very well-drained, alkaline soil, but should face away from midday sun. Keep fairly dry in winter. Propagate by seed in fall or by softwood cuttings in early summer. Susceptible to slug damage. ***P. comosa***, syn. *Phyteuma comosum*, illus. p.366.

PHYSOSTEGIA

Obedient plant

LAMIACEAE/LABIATAE

Genus of summer- to early fall-flowering, erect, deciduous perennials. Fully hardy to frost hardy. Needs sun and fertile, moist but well-drained soil. Propagate by division in spring. ***P. virginiana***. Erect perennial. H 3ft (1m), S 2ft (60cm). Has lance-shaped, sharply toothed, mid-green leaves. In late summer, bears spikes of hooded, 2-lipped, rose-purple flowers with hinged stalks that allow flowers to remain in position once

moved. Z3–9. **var. speciosa** **‘Variegata’** (syn. *P.v. ‘Variegata’*) illus. p.233. **‘Summer Snow’** has pure white flowers. **‘Variegata’** see *P.v. var. speciosa* **‘Variegata’**. **‘Vivid’** illus. p.280.

PHYTEUMA

Horned rampion

CAMPANULACEAE

Genus of early- to midsummer-flowering perennials that are useful for rock gardens. Fully hardy. Needs sun and well-drained soil. Propagate by seed in fall. ***P. comosum***. See *Physoplexis comosa*. ***P. scheuchzeri*** illus. p.342.

PHYTOLACCA

Pokeweed

PHYTOLACCACEAE

Genus of perennials and evergreen shrubs and trees, grown for their overall appearance and decorative but poisonous fruits. Fully hardy to frost tender. Tolerates sun or shade and requires fertile, moist soil. Propagate by seed in fall or spring. ① All parts may cause severe discomfort if ingested; the fruits of *P. americana* may be lethal if eaten. Contact with the sap may irritate skin. ***P. americana***, syn. *P. decandra*. Upright, spreading perennial. H to 12ft (4m), S 3ft (1m). Ovate to lance-shaped, mid-green leaves are tinged purple in fall. Shallowly cup-shaped, sometimes pink-flushed, white-and-green flowers, borne in terminal racemes in summer, are followed by poisonous, rounded, fleshy, blackish-purple berries. Z4–8 ***P. clavigera***. See *P. polyandra*. ***P. decandra***. See *P. americana*. ***P. polyandra***, syn. *P. clavigera*. Stout, upright perennial. H 6ft (2m), S 2ft (60cm). Has brilliant crimson stems and ovate to lance-shaped, mid-green leaves that turn yellow in fall. In late summer, bears clusters of shallowly cup-shaped, pink flowers, followed by poisonous, blackish-purple berries. Z9–11

PICEA

Spruce

PINACEAE

Genus of evergreen conifers with needlelike leaves set on a pronounced peg on the shoots and arranged spirally. Fully hardy to frost hardy. Cones are pendulous and ripen in their first fall; scales are woody and flexible. See also CONIFERS. ***P. abies*** (Common spruce, Norway spruce) illus. p.98. **‘Acrocona’** is a small tree, conical when young, columnar when mature. H 3–10ft (1–3m), S 10–12ft (3–4m). Has pendent branches and blunt, dark green leaves. Even young trees produce abundant cylindrical, deep green cones that turn brown. **‘Clanbrassilliana’**, H and S 20in (50cm), is slow-growing, with a rounded, spreading habit. **‘Inversa’**, H 15–30ft (5–10m), S 6ft (2m), has an erect leader, but pendent side branches. **‘Little Gem’**, H and S 12–20in (30–50cm), has a nest-shaped, central depression caused by spreading branches. **‘Nidiformis’**, H to 5ft (1.5), S 10–12ft (3–4m), also has a “nest” in its center, but is